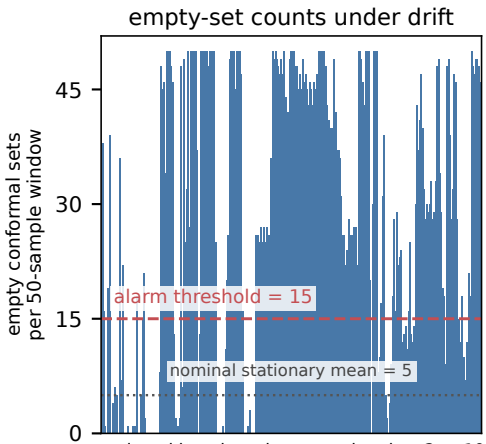
CADENCE a label-free drift alarm with certified pre-adaptation false-alarm control for gas sensor arrays



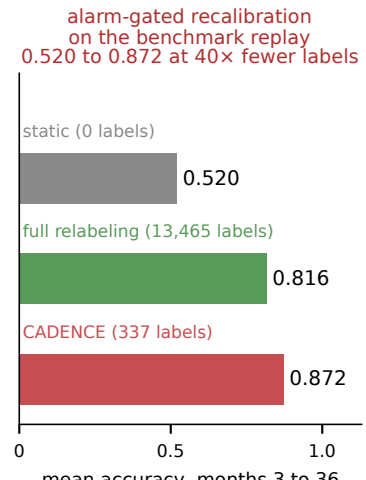
empty prediction set? per sample, one subtraction, one comparison, one conditional increment, and a window-position count

↓

count ≥ 15 in a window raises the alarm false-alarm probability $\leq 10^{-3}$ per window certified under i.i.d. stationarity pre-adaptation epoch

↓

query 10 labels, retrain, resume monitoring



UCI gas sensor drift benchmark replay (10 seeds), alarms in 9 of 9 drifted batches, 0 false alarms in 2,000 stationary windows